even more difficult. Even determining the equations themselves has proved to be so difficult that only approximate versions of them have so far been deduced. String theorists have thereby been limited to finding approximate solutions to approximate equations. After the few years of dramatic progress during the first superstring revolution, physicists found that the approximations being used were inadequate to answer a number of essential questions hindering further developments. With no concrete proposals for going beyond the approximate methods, many physicists working on string theory grew frustrated and returned to their previous lines of research. For those who remained, the late 1980s and early 1990s were trying times. Like a golden treasure securely locked in a safe and visible only through a tiny, tantalizing peephole, the beauty and promise of string theory beckoned, but no one had the key to unlock its power. Long dry spells were periodically punctuated by important discoveries, but it was clear to everyone in the field that new methods with the power to go beyond the previous approximations were required.

Then, in a breathtaking lecture at the Strings 1995 conference held at the University of Southern California—a lecture that stunned a packed audience of the world's top physicists—Edward Witten announced a plan for taking the next step, thereby igniting the "second superstring revolution." String theorists, as of this writing, are working vigorously to sharpen a set of new methods that promise to overcome the theoretical obstacles previously encountered. The difficulties that lie ahead will severely test the technical might of the world's superstring theorists, but the light at the end of the tunnel, although still distant, may finally be becoming visible.

In this chapter and a number that follow, we shall describe the understanding of string theory that emerged from the first superstring revolution and subsequent work prior to the second superstring revolution. From time to time we will indicate new insights stemming from the latter; our discussion of these most recent advances will come in Chapters 12 and 13.

The Greeks' Atoms, Again?

As we mentioned at the outset of this chapter and as illustrated in Figure 1.1, string theory claims that if the presumed point-particles of the standard model could be examined with a precision significantly